

Full deck

Biological Simulations

Stocks, Advice, and Scenarios

From biological advice and “what ifs” to industry resilience
via CREST

Laurence T. Kell ■ BIM Resilience Project ■ July 31, 2026

Outline

- **1. Advice** — mainly conditioned on the *past*
- **2. Resilience** — depends on the *future*, “what ifs”, and does it matter?
- **3. Mackerel** — examples: recruitment, random variability, M shock
- **4. CREST** — fleet outcomes, from diagnosis to prognosis

Resilience means being ready — the future will not be like the past.

Resilience in a Changing World

- **Can't Bounce Back** — productivity and hreference points are *changing*
- **Stay Viable** — not restore an equilibrium

Why it matters

- Past strategies can fail under new regimes
- Fisheries: shifting productivity, distribution, recruitment, ...

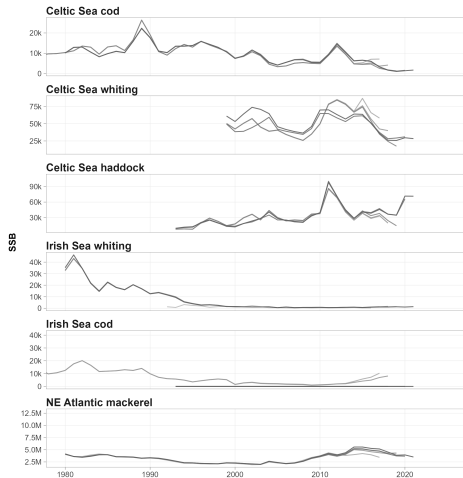
What resilience means

- **Persist** — keep functioning
- **Adapt** — change with conditions
- **Transform** — change the system when needed

1. Advice — annual updates

Update: advice is based on

- Adding last year's catches and surveys, then re-running the model
- Series extends one year; structure and reference points stay the same
- Terminal year moves

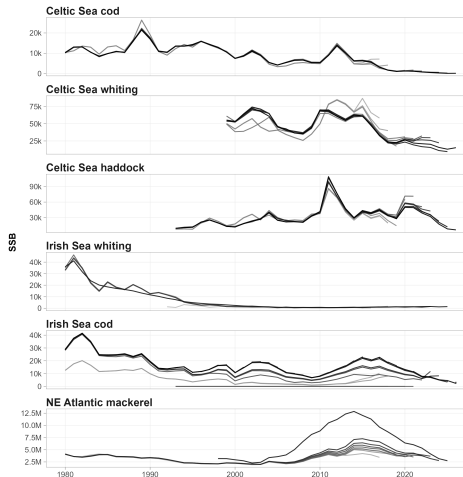


SSB by stock — update peels 2016–2021

1. Advice — benchmarks change the story

Benchmark: periodic reset

- Full peer review of assumptions, data and reference points
- B_{trigger} , B_{lim} and F_{MSY} change
- Productivity regimes change them further
- **Resilience:** even the past may change

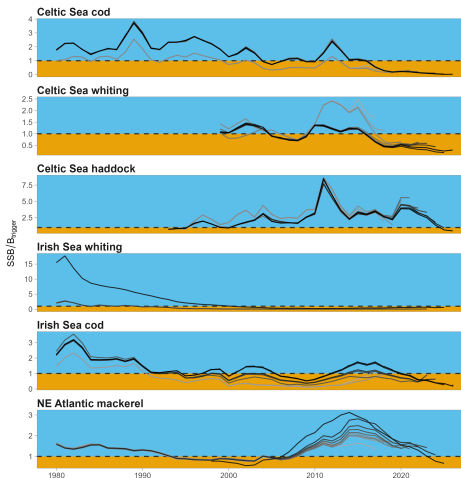


SSB by stock — assessment years 2016–2026

1. Advice — status relative to B_{trigger}

Management is relative to reference points:

- $SSB/B_{\text{trigger}} = 1$
 - > 1 — carry on
 - < 1 — reduce F , catch and effort
- Crossing the threshold flips status and advice
- Benchmark can change B_{trigger} and flip the story



SSB/B_{trigger} — blue above 1, orange below

2. Scenarios — what ifs

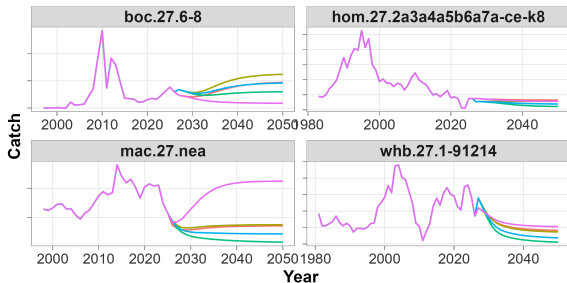
Examples

- Projections under **fixed** fishing effort, or MSE for ICES AR
- Common questions:
 - Status-quo F with recent productivity
 - Same F with poorer recruitment (75%, 50%)
 - F_{MSY} for recent recruitment *regime*

2. Scenarios — pelagics

Pelagics: boarfish, western horse mackerel, NEA mackerel, blue whiting

- Status-quo, recruitment regimes
- **Resilience:** under current fishing, does catch get better, stay the same, or get worse?

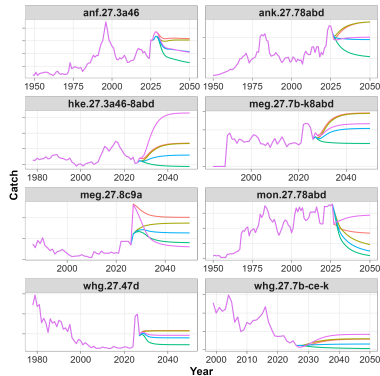


Projected catch by scenario

2. Scenarios — demersal

Demersal: anglerfishes, northern hake, megrims, whittings

- Status-quo, recruitment regimes
- **Resilience:** under current fishing, does catch get better, stay the same, or get worse?



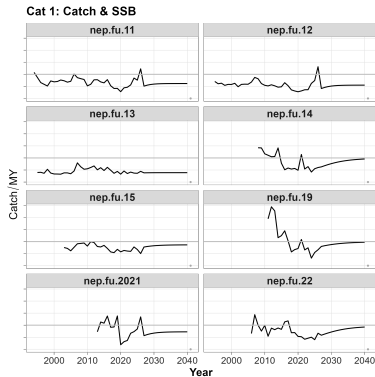
Projected catch by scenario

2. Scenarios — Nephrops

Nephrops: functional units OMs
(JABBA / biodyn)

- **Scenarios:**

- Status-quo harvest; same harvest with process error $\times 0.75$ / 0.50
- **Resilience:** spatial squeeze, effort $\neq F$?



Projected catch by scenario

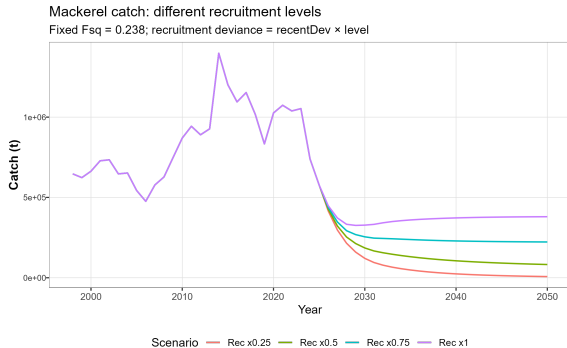
3. Mackerel — example

“What if” based on current F

- Stresses
 - Lower recruitment (productivity)
 - One-off natural-mortality shock (a flip, not a trend)
 - Year-to-year recruitment noise (the spread fleets live with)
- Biological trajectories are what CREST carries through to fleet values and effort

3. Mackerel — recruitment levels

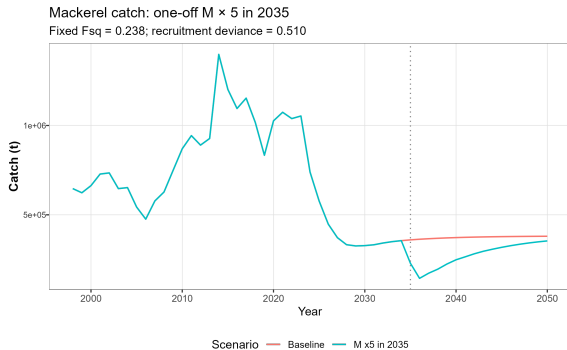
- Fixed status-quo F
- Recruitment deviance scaled relative to the recent residual mean:
 - $\times 1$, $\times 0.75$, $\times 0.5$, $\times 0.25$
- Question: if effort stays as now, how far does catch fall when productivity is poorer?



Catch under recruitment multipliers

3. Mackerel — one-off M shock

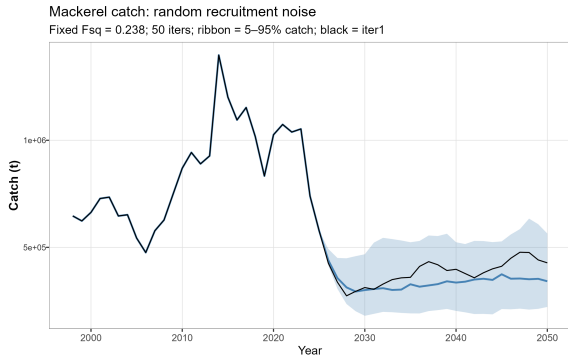
- Same status-quo F and baseline recruitment deviance
- **Shock:** natural mortality $\times 5$ for a single year (2035), then back to baseline M
- A pulse — the kind of flip a smooth forecast misses
- Compare shocked catch path with the no-shock baseline



Catch: baseline vs $M \times 5$ in 2035

3. Mackerel — random recruitment

- Year-to-year noise around recent recruitment
- Recent F
- Ribbon = 5–95% of **catch** paths; black = one example year sequence
- **Resilience**: what fleets have to live with — not a mean trend

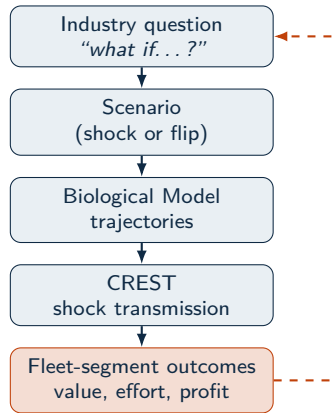


Stochastic catch: median, band, example path

4. CREST — not an answering machine

CREST: Capacity, Resilience and Economic Shock Transmission

- **Industry asks** “*what if . . . ?*”
- That becomes a scenario — then: is the fleet resilient, or should it be?
- Carries biological, regulatory or market shocks through to fleet value, effort and profit
- Sits between stock assessment and a whole-economy model
- Framed around **shocks or flips**, not trend extrapolation



Outcomes reframe the next question — resilient, or should it be?

5. Expert — which uncertainties matter?

- Scoping *which* shocks are worth testing is a formal step — structured elicitation, not only an analyst's list
- Precedent (Eastern Atlantic bluefin, ICCAT/GBYP): uncertainties scored for *importance*, *knowledge* and *representation* (Leach et al., 2014)
- That prioritisation feeds CREST scenario design
- Policy response: insurance / mutual funds to mediate revenue shocks (Mumford et al., 2009)

Elicitation dimensions

- Importance — impact on objectives
- Knowledge — reducible by research?
- Representation — in the assessment?

Closing thread

- Advice conditions on the past
- Scenarios probe flips ahead
- CREST asks: is the fleet ready?

References

- Kell, L.T. (2026). *Simulation of future TACs*, draft chapter, BIM Resilience report.
- Mumford, J.D., Leach, A.W., Levontin, P., Kell, L.T. (2009). Insurance mechanisms to mediate economic risks in marine fisheries. *ICES J. Mar. Sci.*, 66, 950–959.
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- Leach, A.W. et al. (2014). Identification and prioritization of uncertainties for management of Eastern Atlantic bluefin tuna. *Marine Policy*, 48, 84–92.
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